

SMARLA: A **Safety Monitoring Approach for Deep Reinforcement Learning Agents**

Amirhossein Zolfagharian, Manel Abdellatif, Lionel C. Briand, Ramesh S.

IEEE Transactions on Software Engineering (IEEE TSE), 2025.

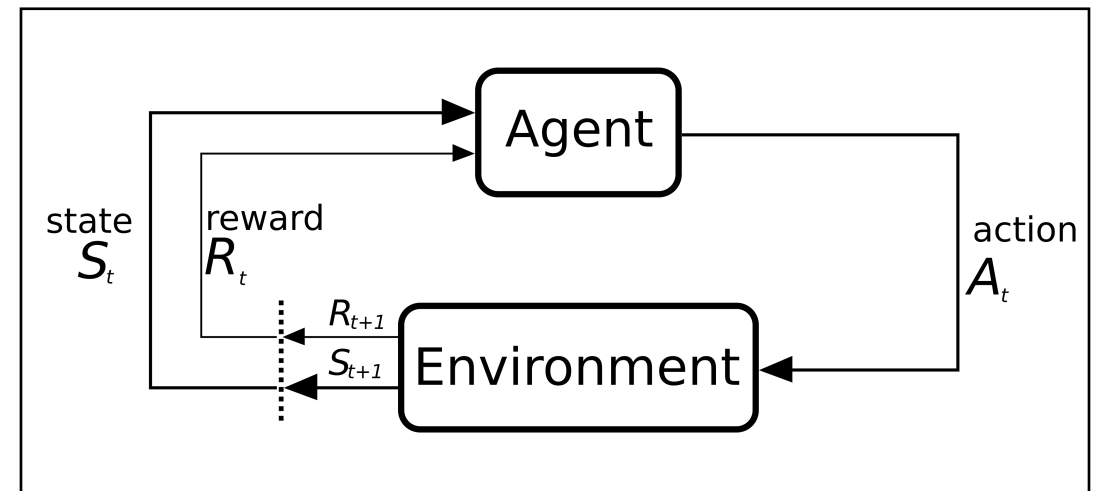
JOÃO EDUARDO MONTANDON

What is Reinforcement Learning (RL)

Involves **training an agent through interactions** with its environment.

The **goal** of the agent is to **maximize its rewards** by trying new **actions to learn** more about the environment.

Extensively used in robotics, where environment is generally diverse and unknown.



Deep Reinforcement Learning (DRL)

Extension of RL where optimal **policies are learned with Deep Neural Networks** (DNNs).

Advantage: allow agents to be trained directly from raw data.



Challenges with DRL

As with typical ML component, the behavior emerges from the data the agent is being exposed on.

- A. DRL agents often operate in **environments** with **extremely large** state spaces.
- B. They also adapt their behavior based on interactions with the **environment**, which is completely **uncertain**.

AARIAN MARSHALL

GEAR APR 26, 2024 3:38 PM

Tesla Autopilot Was Uniquely Risky—and May Still Be

In an investigative report into crashes and deaths associated with Tesla Autopilot, federal regulators concluded that the system lacked standard protections.



PHOTOGRAPH: PATRICK PLEUL/GETTY IMAGES

BREAKING | BUSINESS > BREAKING NEWS

Tesla Faces Investigation Over Reports Full-Self Driving Runs Red Lights

By [Zachary Folk](#), Forbes Staff. I cover breaking news.

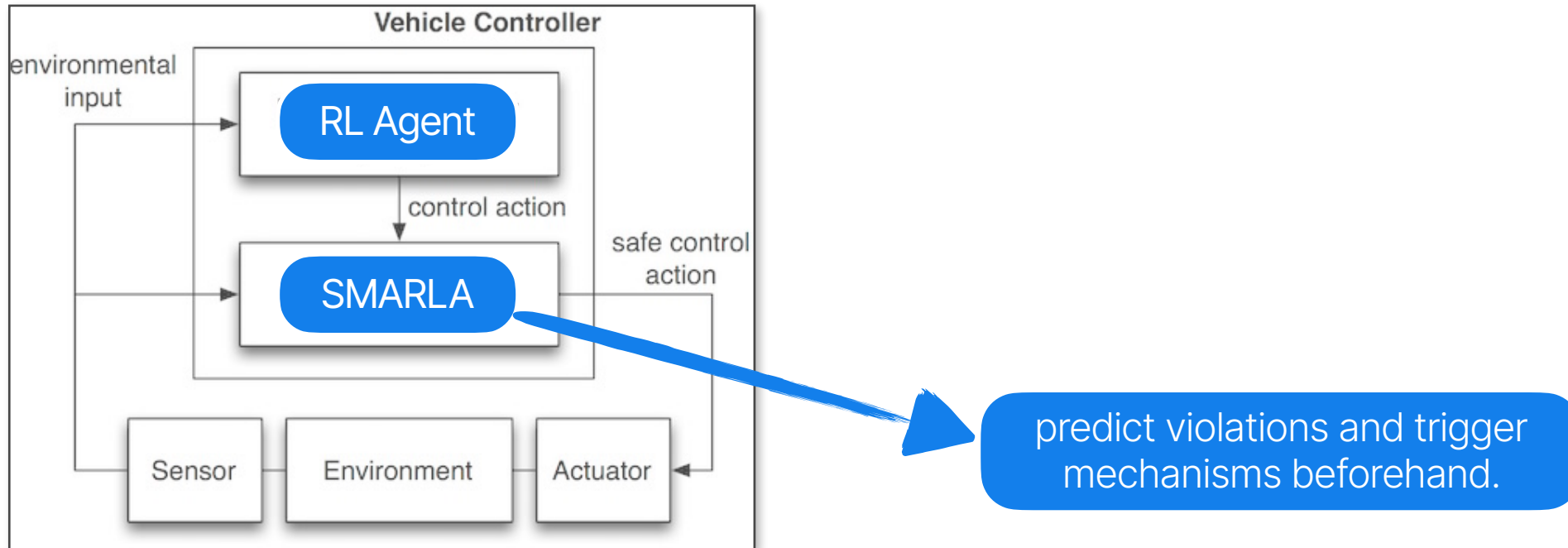
[Follow Author](#)

Published Oct 09, 2025, 10:17am EDT

How to **ensure [safety] requirements will not be violated?**

The Proposed Approach

SMARLA: a **safety monitoring** approach that uses **Machine Learning** to predict safety violations during the execution of agents.

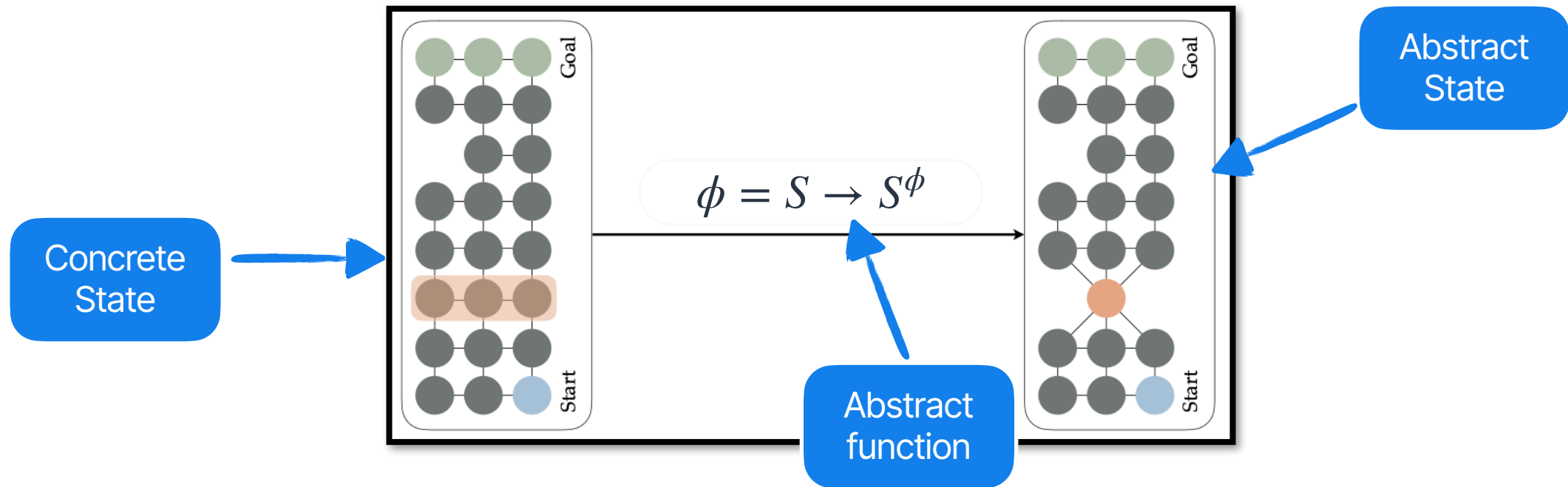


Some Definitions

State Abstraction

A technique to **reduce the state space** and thus enhance the learnability of the ML model.

Generally performed with clustering techniques.

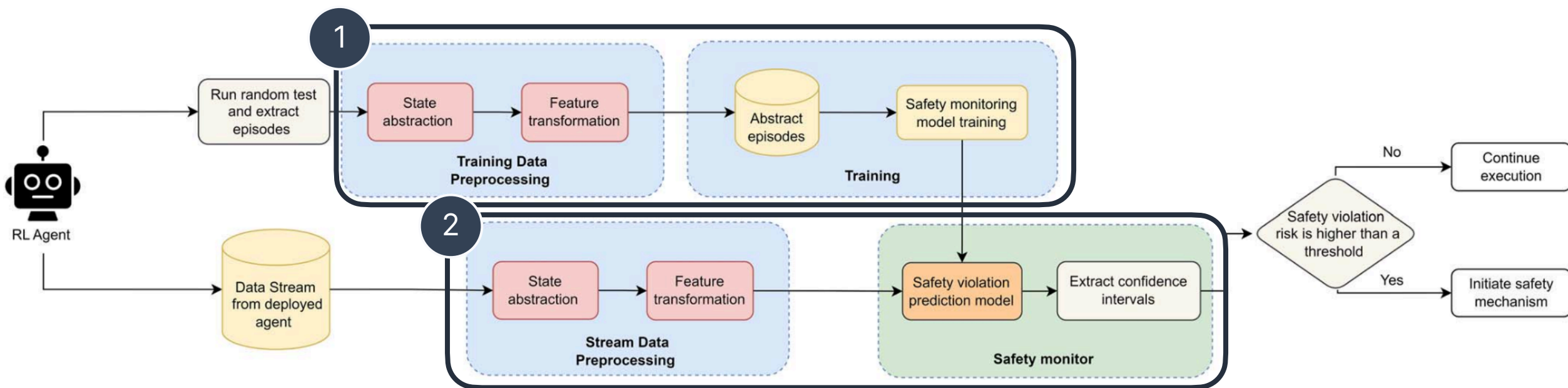


Data Structure

- **State:** a pair $\langle s_i, a_i \rangle$, where $s_i \in S$ and $a_i \in A$.
- **Episode:** a finite sequence of state-action pairs $\langle s_i, a_i \rangle$, representing the execution of the RL Agent.
- **Unsafe state:** a state in which at least one safety requirement is violated, e.g., car collision.
- **Unsafe episode:** an episode that contains an unsafe state.

SMARLA

Overview



1. Training

Data collection

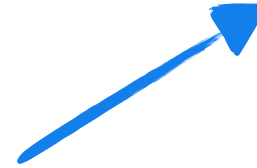
- Data collected by randomly simulating the RL agent on different episodes.
- Episodes are labeled as unsafe if the last state is unsafe.

1. Training

State Abstraction

A. State abstraction is built after collecting and labelling all training episodes.

```
def state_abstraction(episode):  
    e_abs = []  
    for state s in episode.states:  
        s_abs = match_abs_state(s)  
        if s_abs then:  
            e_abs[s_abs] = 1  
        else:  
            e_abs[s_abs] = 0  
    return e_abs
```



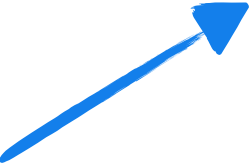
	S_1^ϕ	S_2^ϕ	$\dots$	S_j^ϕ	$\dots$	S_{n-1}^ϕ	S_n^ϕ
$episode_i$	1	0	$\dots$	1	$\dots$	1	0

1. Training

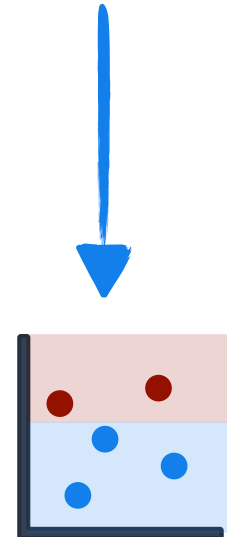
State Abstraction

B. Random Forest classifier is trained on the abstract episodes.

```
def state_abstraction(episode):  
    e_abs = []  
    for state s in episode.states:  
        s_abs = match_abs_state(s)  
        if s_abs then:  
            e_abs[s_abs] = 1  
        else:  
            e_abs[s_abs] = 0  
    return e_abs
```



	S_1^ϕ	S_2^ϕ	$\dots$	S_j^ϕ	$\dots$	S_{n-1}^ϕ	S_n^ϕ
$episode_i$	1	0	$\dots$	1	$\dots$	1	0



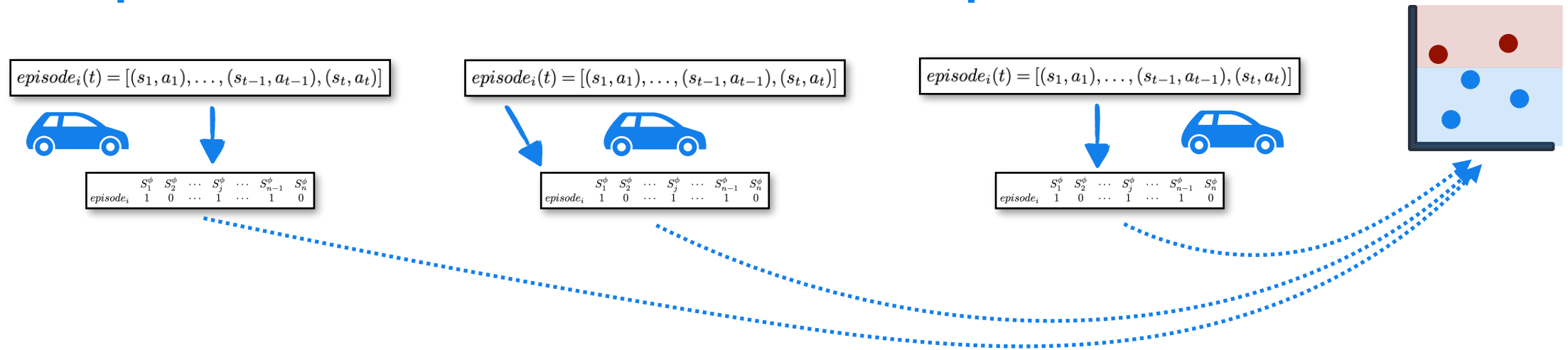
2. Operation

Pre-processing

Streamed episodes can be different in length.

We apply pre-processing step to fit the size of streamed episode to the number of abstract states defined in training.

The **episode is fed** into the model **at each time step t** .



If the transformed abstract **state is unseen**, it is **discarded**.

2. Operation

Prediction

SMARLA supports three different decision criteria.

- A. **Upper Bound:** $P(t) + CI(t) \geq 0.5$
Favors early detection, but increase FPs.
- B. **Output:** $P(t) \geq 0.5$
Balanced, accuracy is more relevant.
- C. **Lower Bound:** $P(t) - CI(t) \geq 0.5$
Avoids FPs, late detection is not severe.

Empirical Evaluation

Research Questions

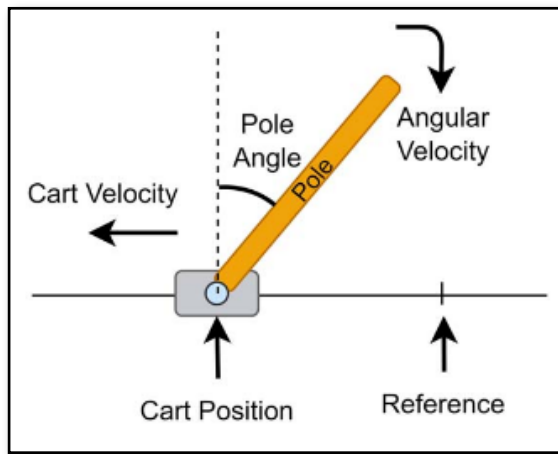
RQ1: How accurately and early can we predict safety violations?

RQ2: How can the safety monitor determine when to trust the prediction of safety violations?

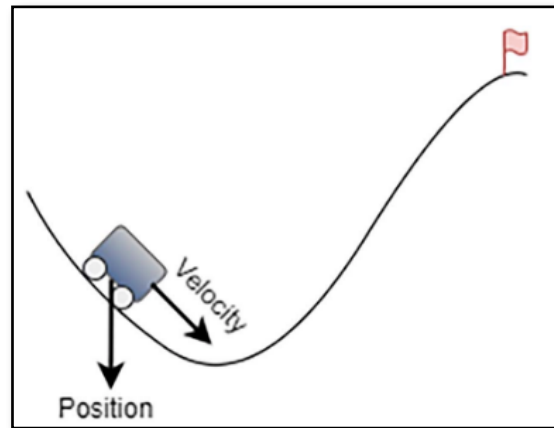
RQ3: What is the effect of the prediction threshold?

RQ4: What is the effect of the abstraction level applied on the states?

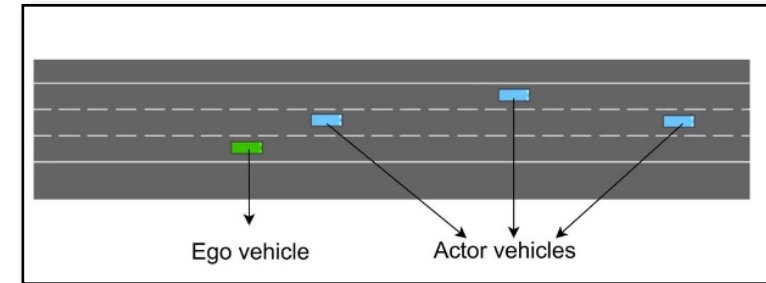
Case Studies



Cart-Pole



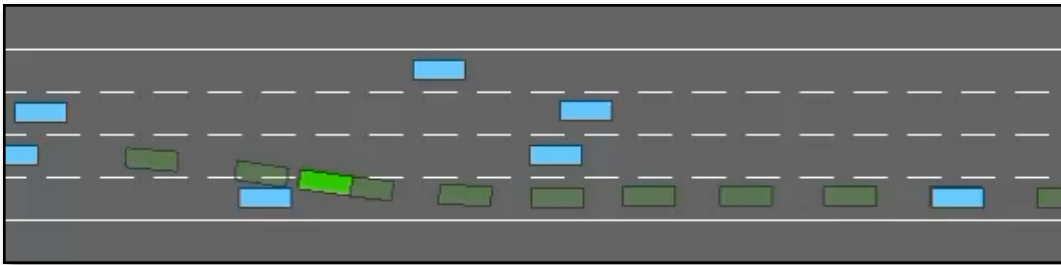
Mountain-Car



Highway Driving

Case Studies: Highway Driving

An **RL agent is responsible for driving** a car on a highway with three lanes.



- A. The **goal** is to **drive in high speed without colliding**.
- B. Five actions: move right, move left, accelerating, idle, braking.
- C. Episodes have **30 time steps**.

Training Details

A. RL agent

- Trained on 70,000 time steps.
- On average, car drives safely for 29 time steps.

B. Safety monitor (SMARLA)

- Sampled through random execution of RL agent.
- Trained on 4,000 episodes; 222 unsafe.
- Abstraction level: $0.2 \Rightarrow 115,730 \ggg 136$

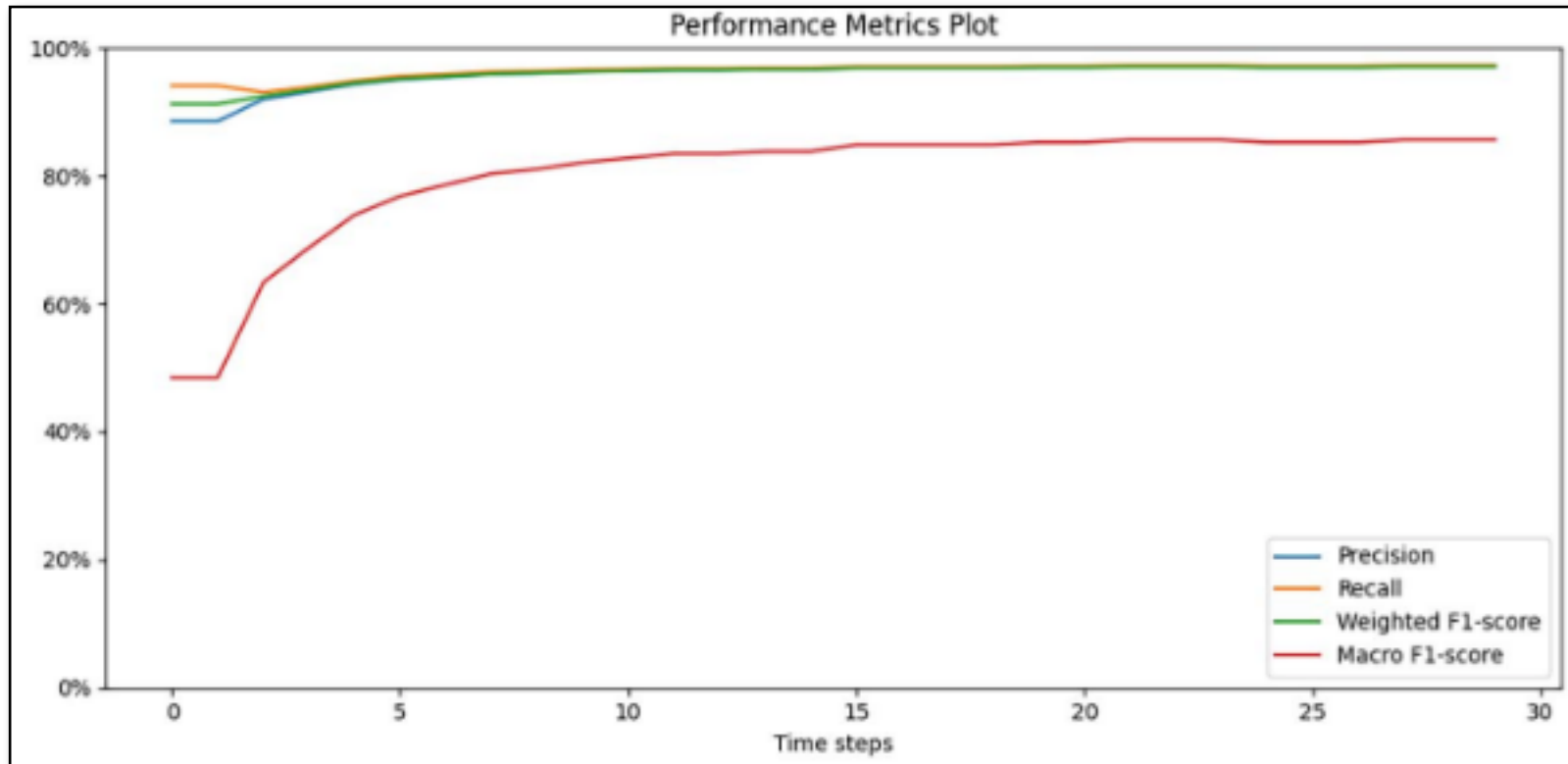
Experiment Ground Truth

- 1000 random executions of RL agent.
Each episode was labelled as safe or unsafe after execution.
- SMARLA monitored each execution for each time step.
- $Up(t)$ prediction policy was used, favoring early detection.
- Metrics: Precision, Recall, Weighted F1, Macro F1.

Results

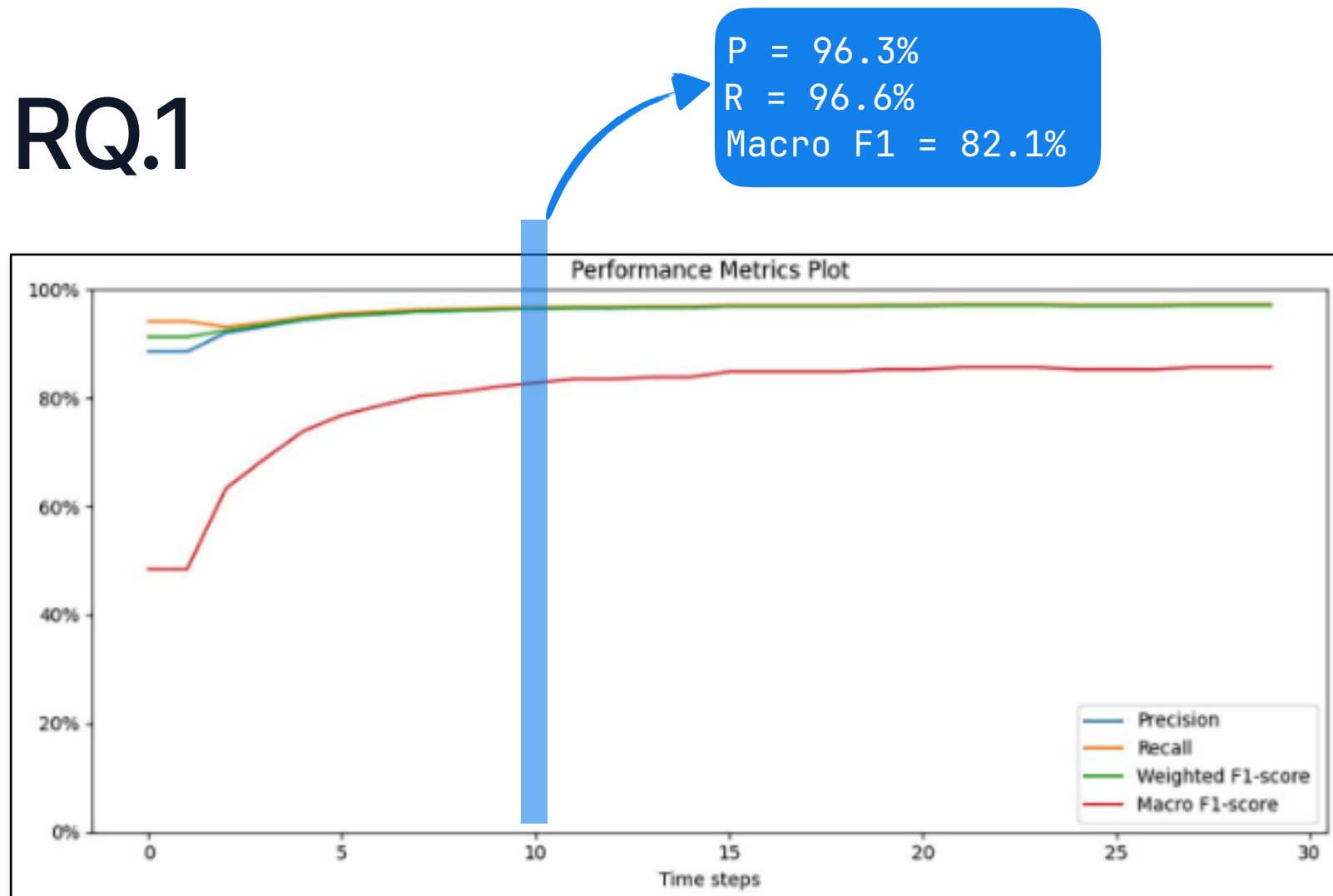
Highway Driving case study

RQ.1



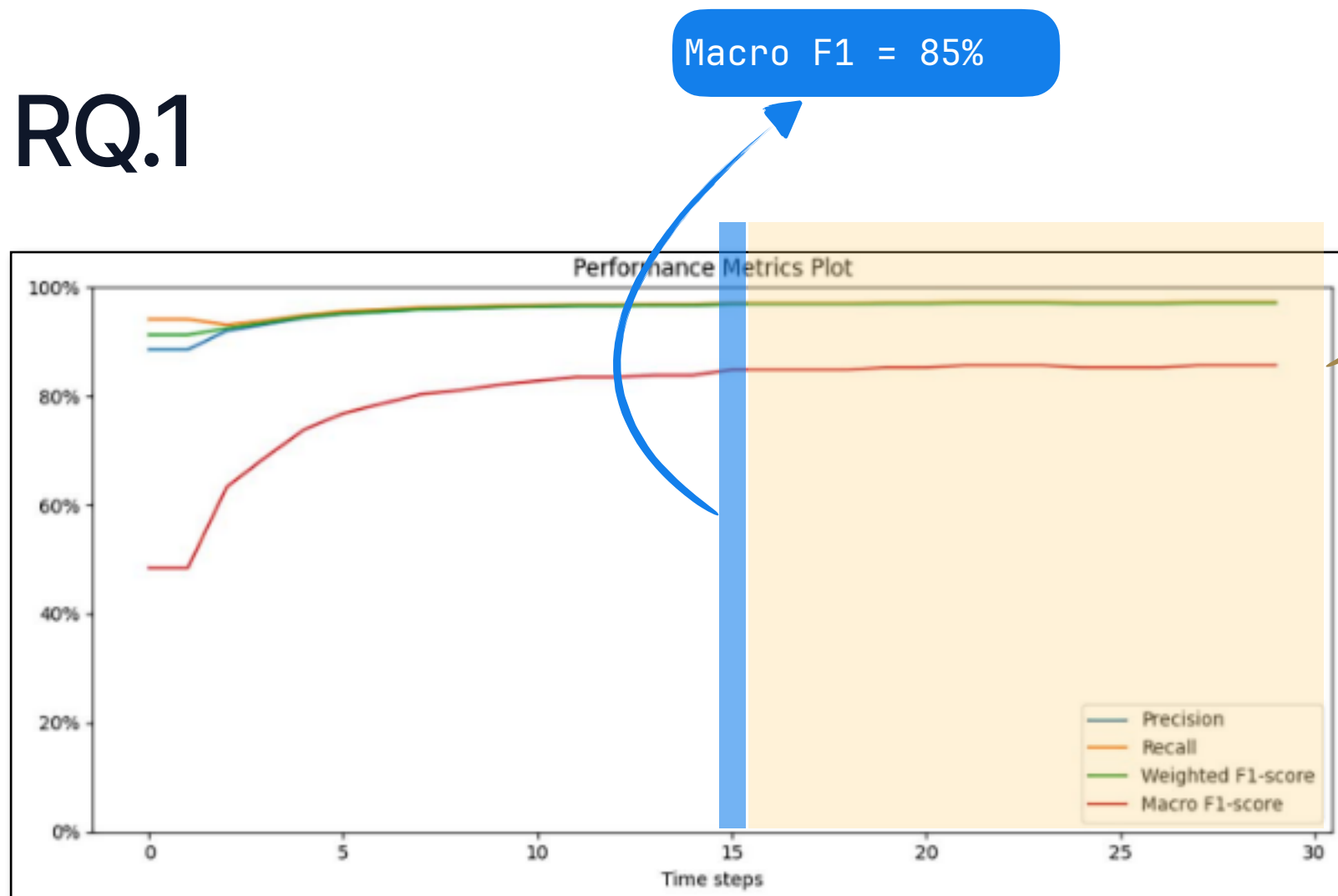
SMARLA correctly detected 40/59 safety violations.

RQ.1



SMARLA correctly detected 40/59 safety violations.

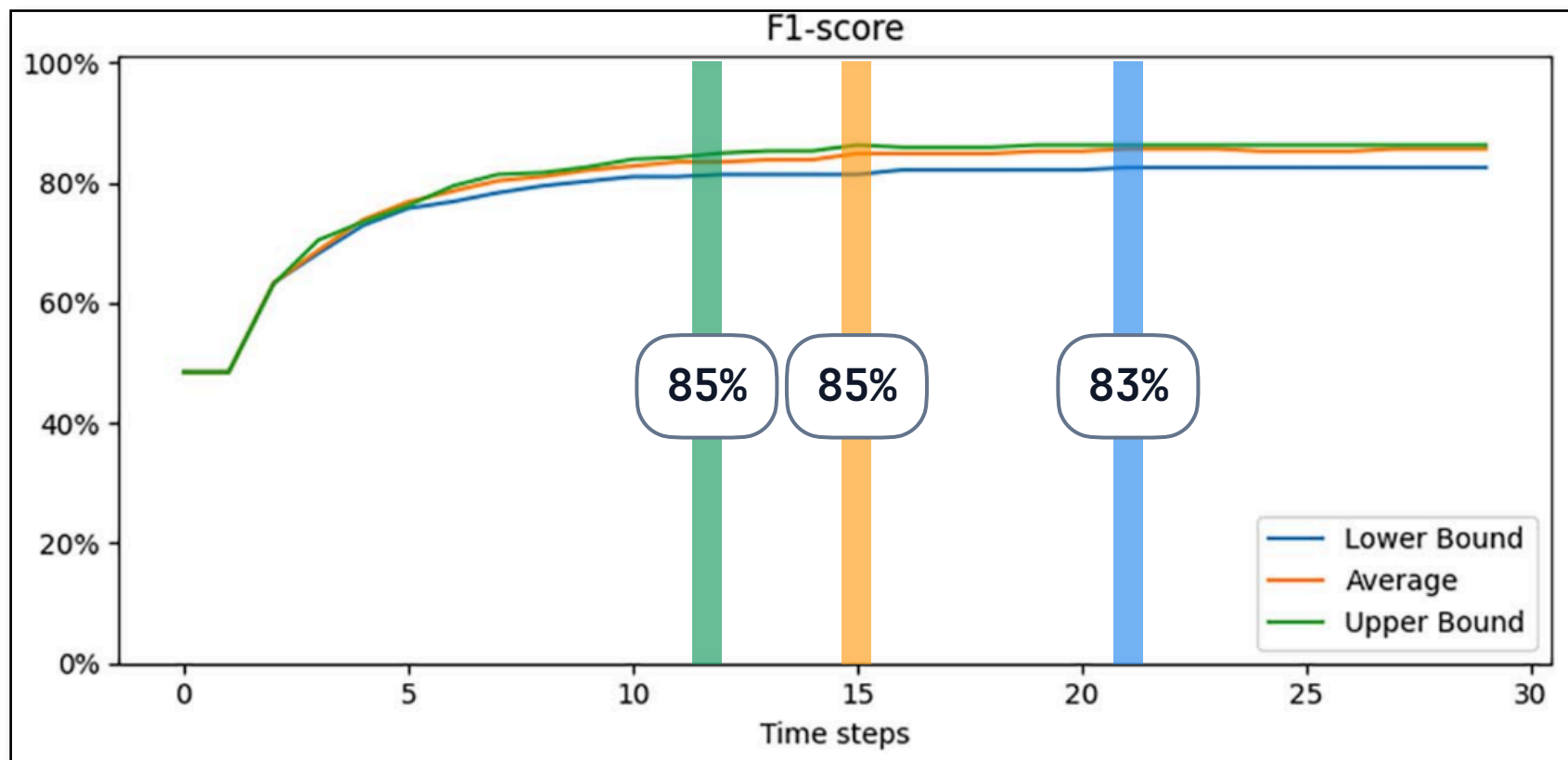
RQ.1



On average, SMARLA anticipated violations by 48%

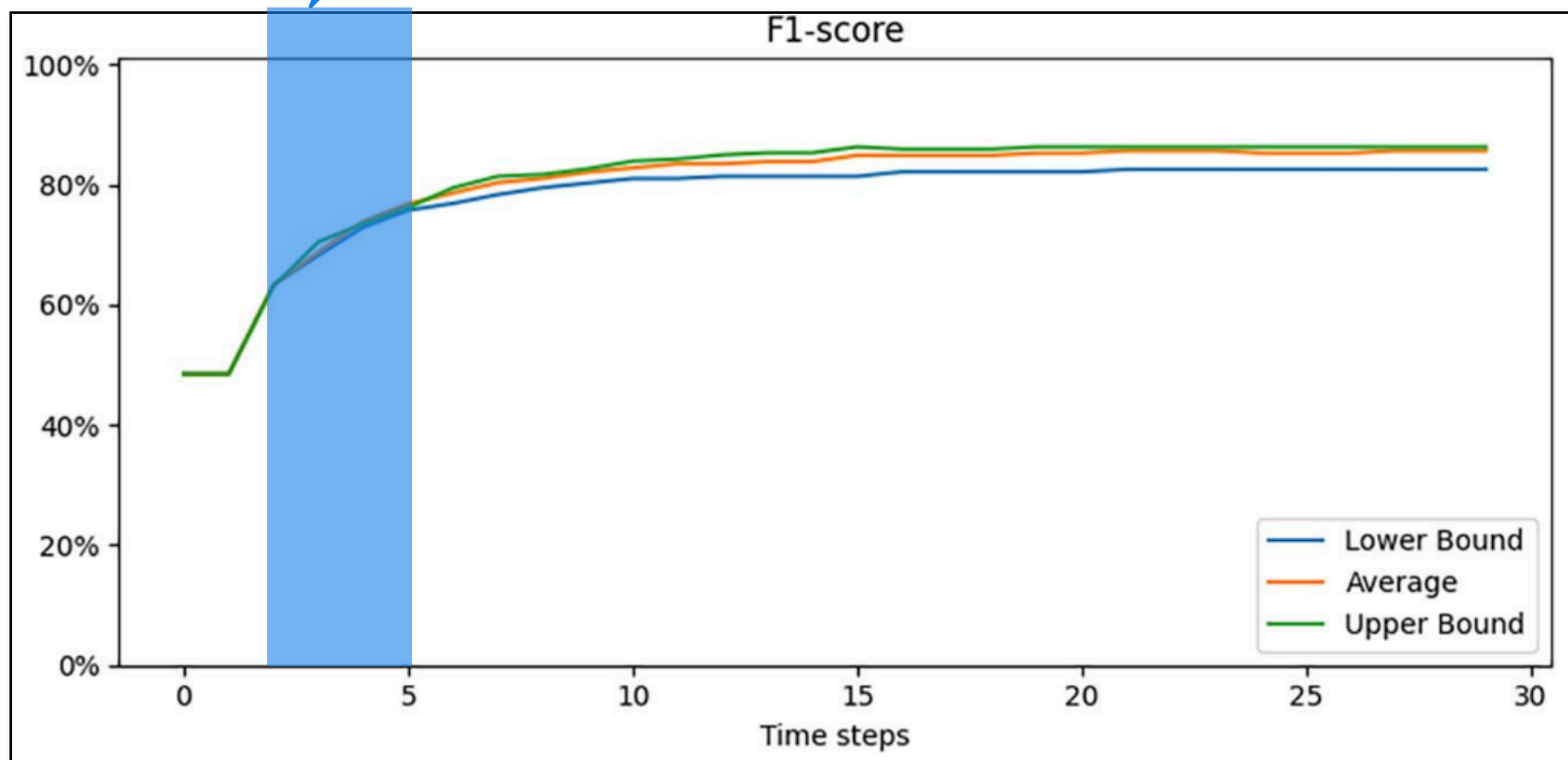
SMARLA correctly detected 40/59 safety violations.

RQ.2



RQ.2

► Sharp increase due to cold start problem



RQ.2

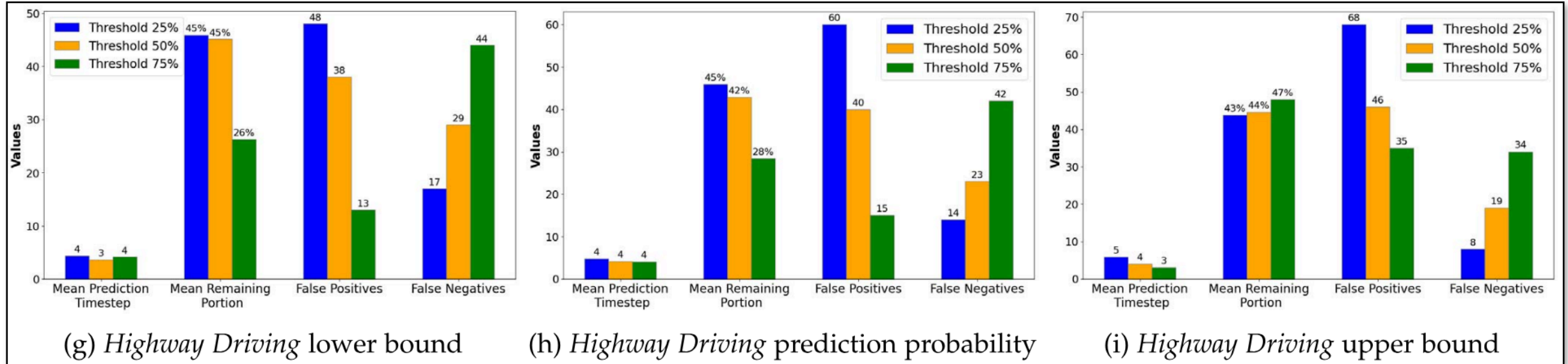
	Stabilized step	FPS	FPS rate
Upper Bound	12	46	4.8%
Output	15	40	4.2%
Lower Bound	21	38	4%

Stabilized 09 steps earlier

Only 0.8% more FPS

The diagram illustrates the performance metrics for three scenarios: Upper Bound, Output, and Lower Bound. The 'Stabilized step' column shows values 12, 15, and 21 respectively. The 'FPS' column shows 46, 40, and 38. The 'FPS rate' column shows 4.8%, 4.2%, and 4%. Annotations highlight that the 'Stabilized step' is 09 steps earlier for the Upper Bound and that the FPS rate is only 0.8% more for the Output scenario compared to the Lower Bound.

RQ.3

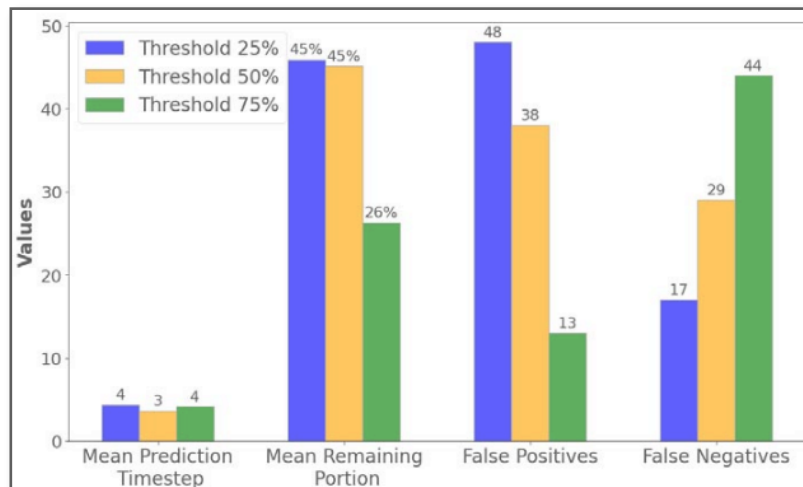


25% $\Rightarrow$ earlier detection, but at a cost of more FPs.

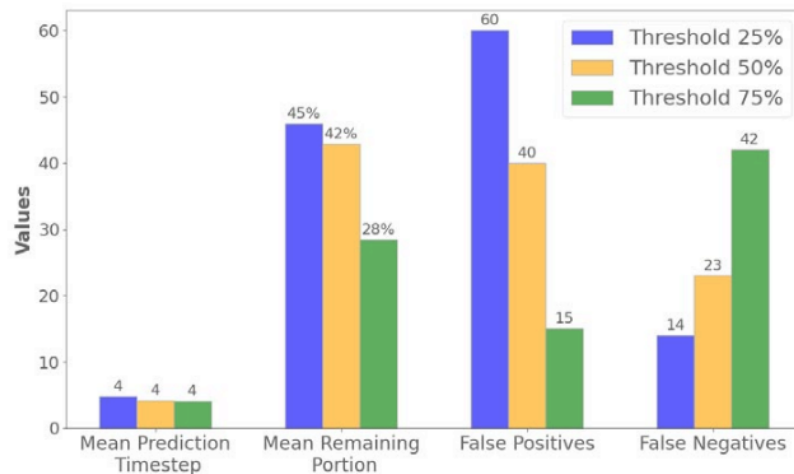
50% $\Rightarrow$ balanced setting; early detection with reasonable FPs.

75% $\Rightarrow$ reduced FPs, but detection happens in the last third.

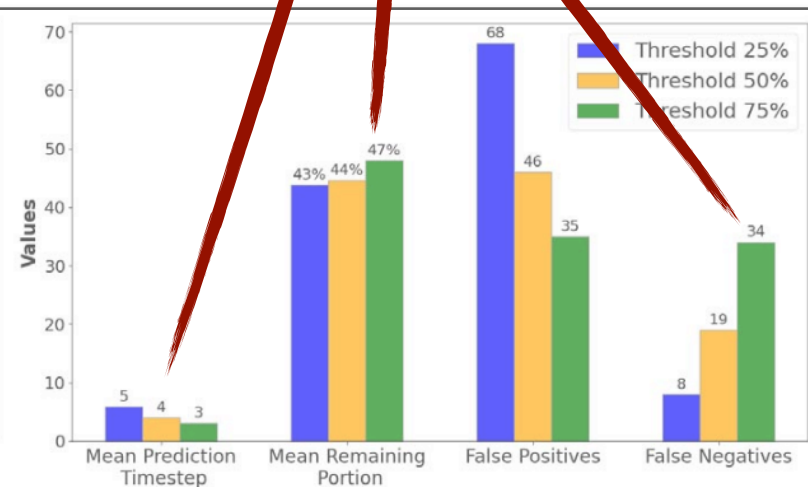
RQ.3: Don't do that!



(g) Highway Driving lower bound



(h) Highway Driving prediction probability

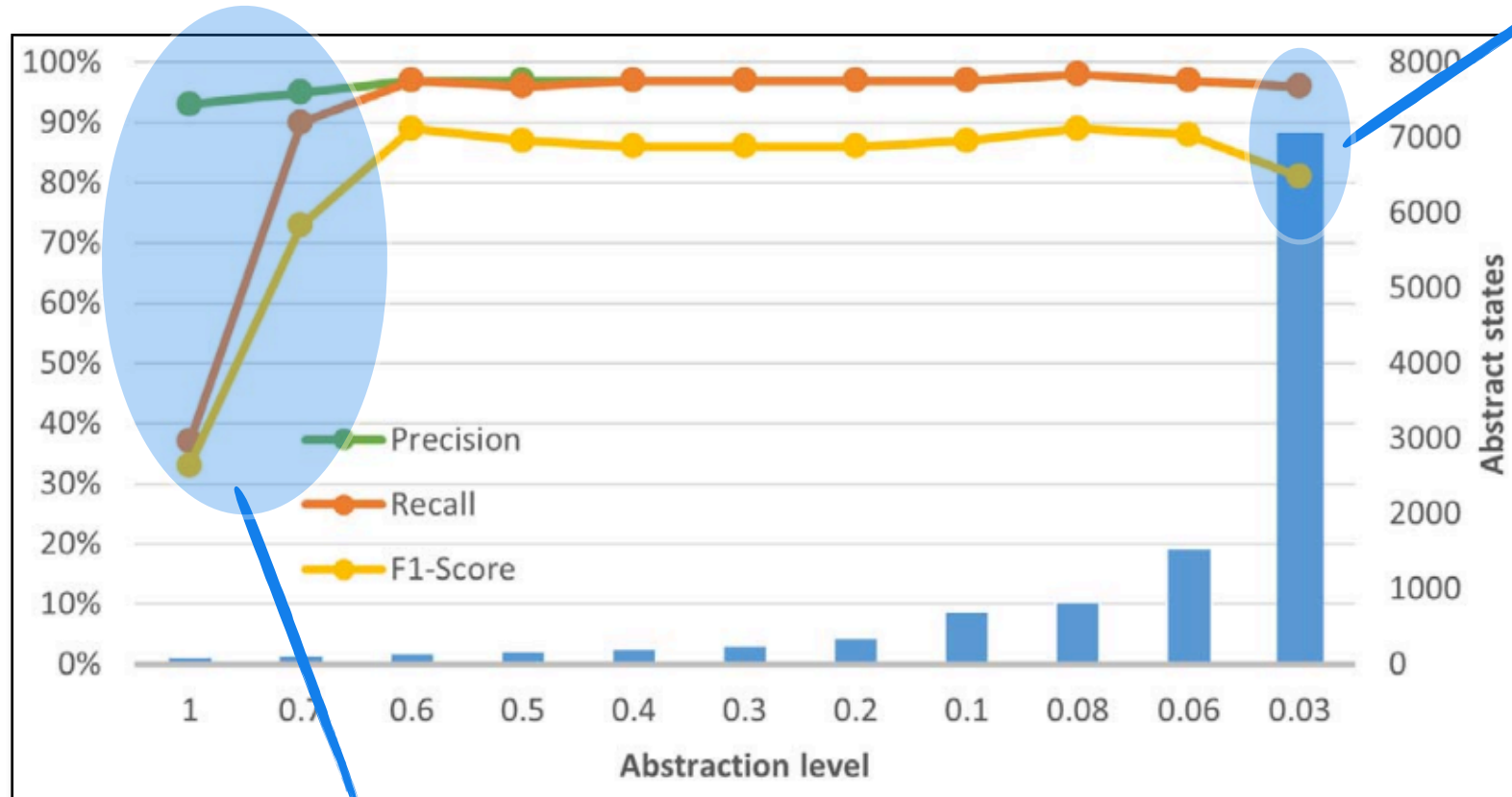


(i) Highway Driving upper bound

Different scales

Incomparable measures

RQ.4

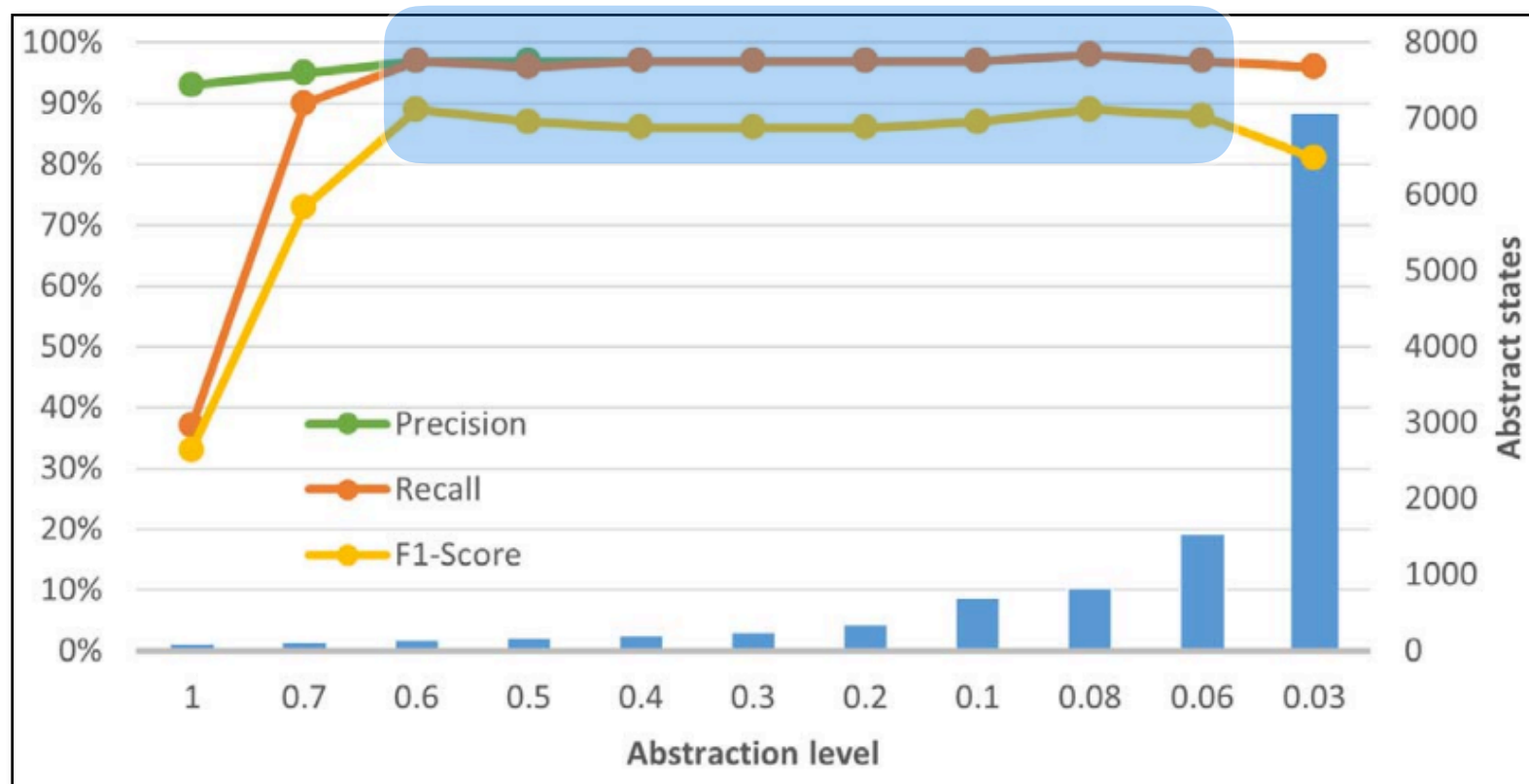


Too many = noise

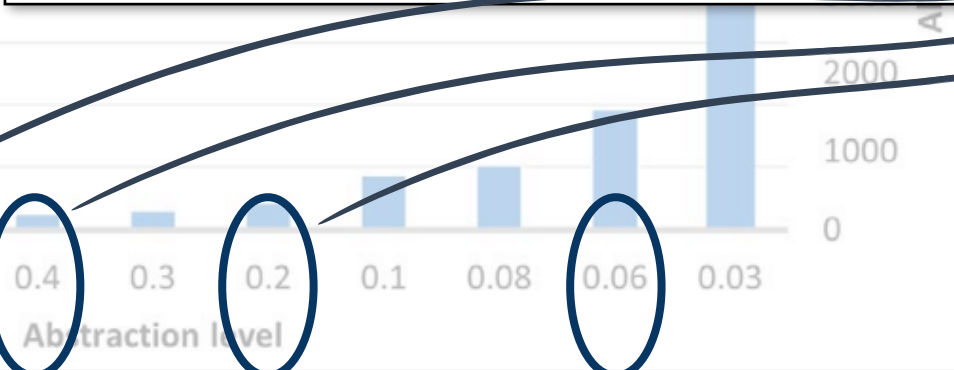
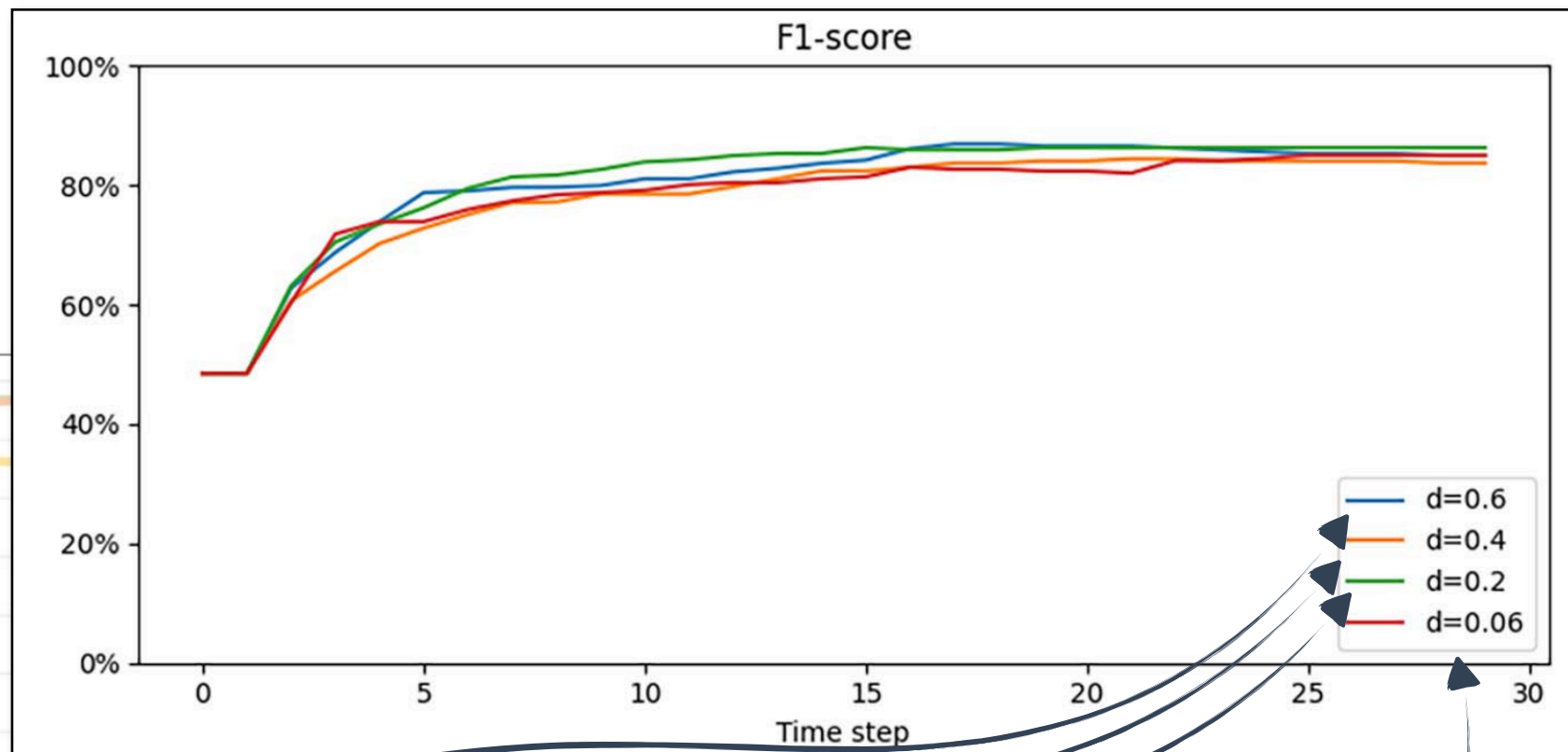
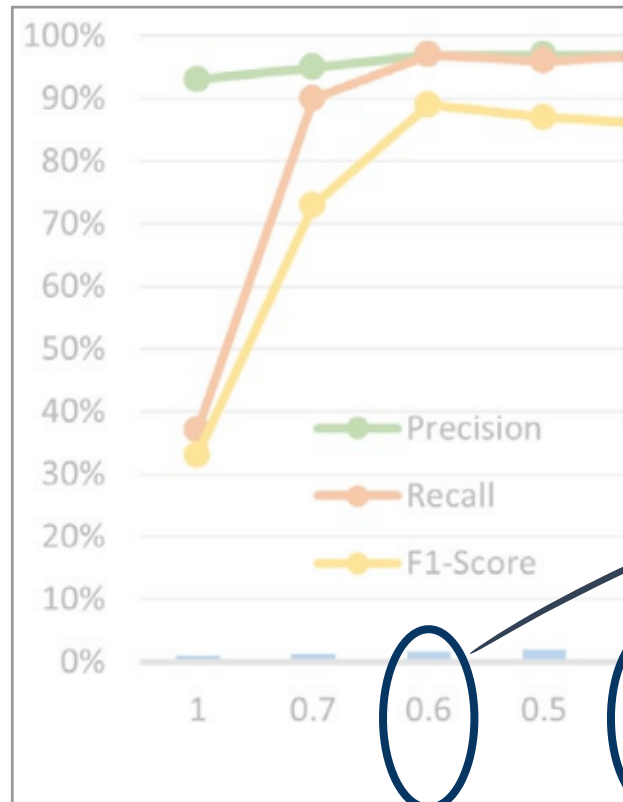
Too few = information loss

RQ.4

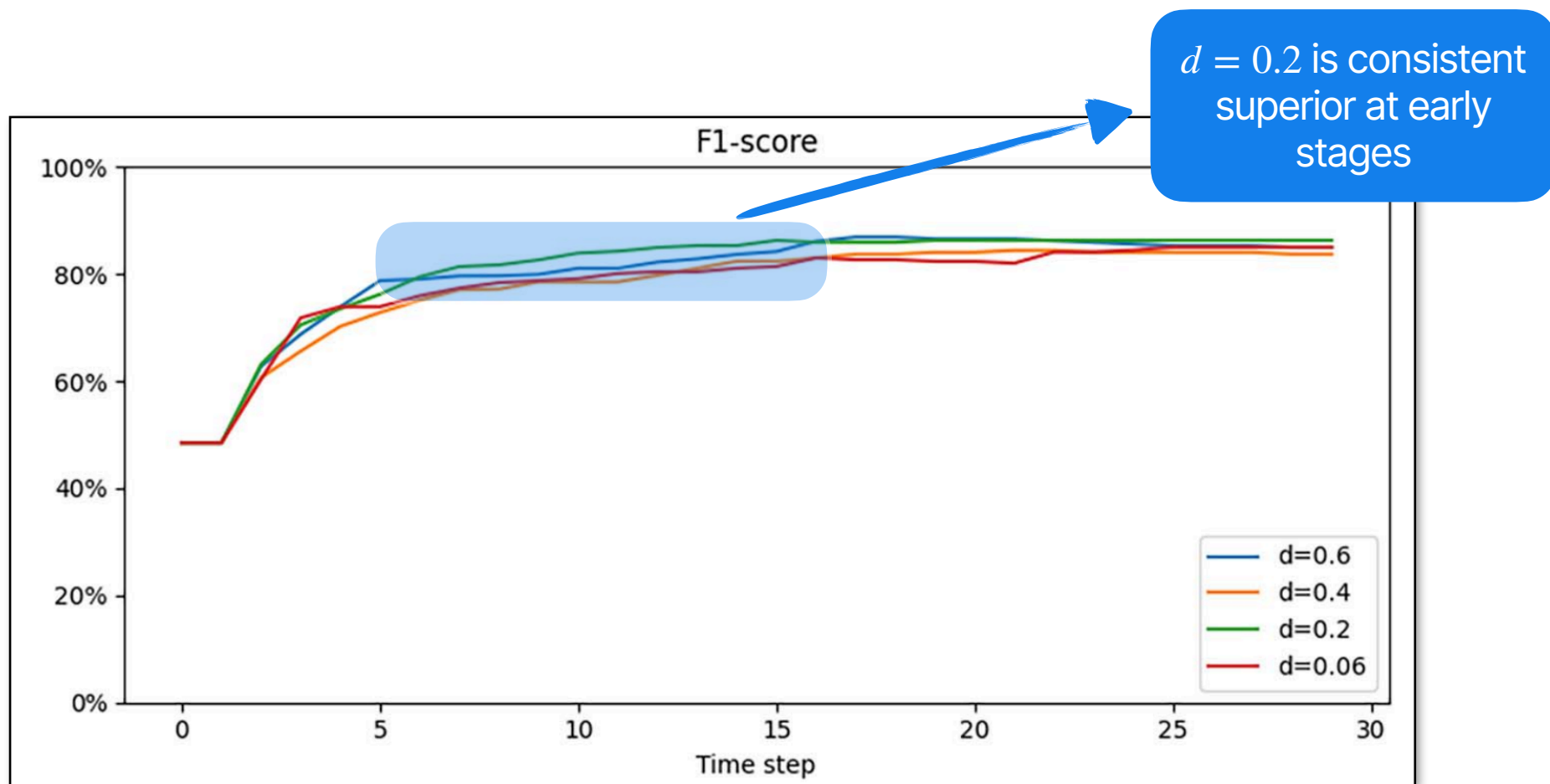
Optimal range



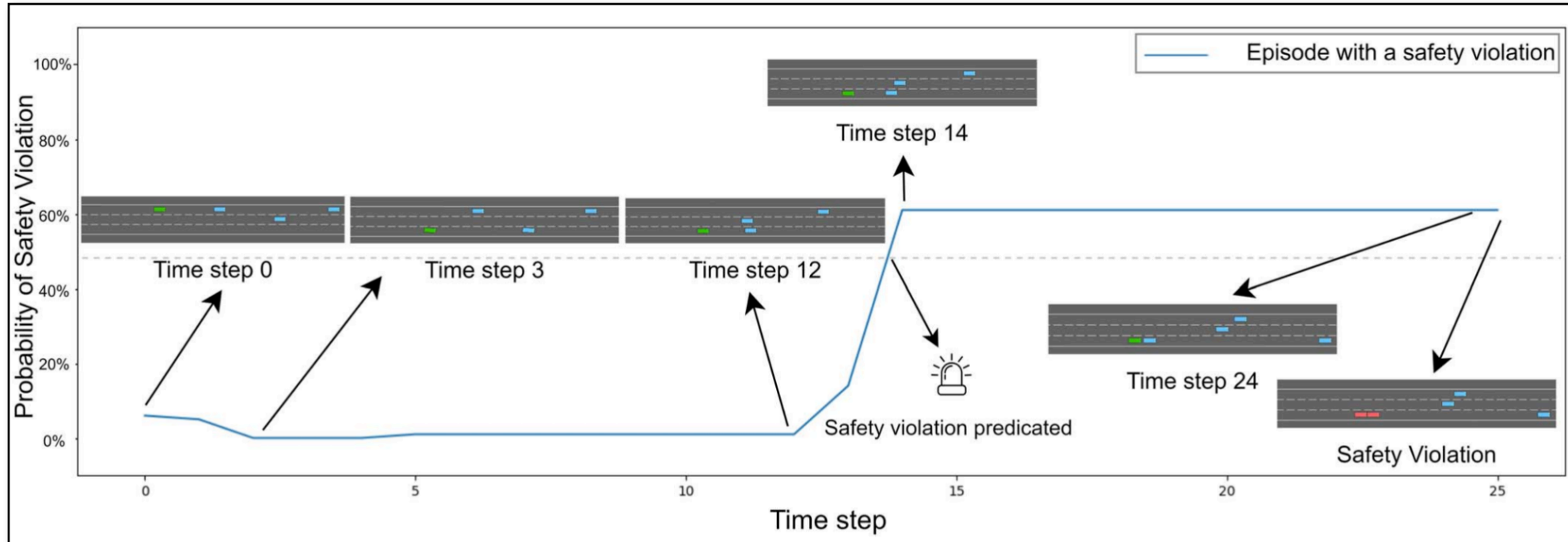
RQ.4



RQ.4



A Concrete Example



Violation predicted **9 steps before**.

Final Thoughts

- SMARLA is a black-box safety monitoring approach, i.e., depends only on the agent input/output.
- It is a data-based solution, 100% coverage is not guaranteed.
 - State abstraction can help generalizing it, though.
- Trade-off between early prediction and FPs are directly related to the task.

joao@dcc.ufmg.br

Thanks!